

Module 13: Token Bus (802.4), Token Ring (802.5) & FDDI Architecture

Unit 2: Data Link Layer & MAC Sublayer

Deterministic LANs & Optical Self-Healing

Comprehensive architectural study of IEEE 802.5 Token Ring, MSAU star-ring wiring, Active Monitor recovery, and FDDI dual counter-rotating ring self-healing loopback mechanics.

Token Ring (IEEE 802.5) & FDDI Dual Counter-Rotating Self-Healing Architecture

1. IEEE 802.5 Token Ring & MSAU

Token Format (3B):
SD (1B) + AC (1B) + ED (1B)

Active Monitor (S1):

- Prevents orphaned frames.
- Regenerates lost token.
- Sets Monitor bit in AC.

Speed: 4 / 16 Mbps

Early Token Release (ETR):
In 16 Mbps mode, station releases token immediately after transmitting frame without waiting for frame header to travel around the entire ring.

2. FDDI Dual Counter-Rotating Self-Healing Ring

Self-Healing Wrap:
When cable snaps, stations B and C wrap primary ring into secondary ring!
Forms 1 Big Single Ring
Network stays 100% alive!
Span: 200 km, 1000 Stns

Timed Token Protocol in FDDI:
TTRT (Target Token Rotation Time) negotiated by stations.
Token Holding Time: $THT = TTRT - TRT$ (Guarantees Real-time voice bandwidth!).

Comparative Architecture: Token Bus (802.4) vs Token Ring (802.5) vs FDDI

Token Bus (802.4):	Physical Bus, Logical Ring. Stations ordered by descending numerical address. Complex ring maintenance.
Token Ring (802.5):	Physical Star / Logical Ring via MSAU. Token captured by setting T bit in Access Control byte. Monitored by Active Monitor.
FDDI (ANSI X3T9.5):	Dual Counter-Rotating Fiber Rings. 100 Mbps. 4B/5B block coding. Self-healing wrap-around on fiber rupture.
Efficiency Law:	$FDDI \text{ Token Holding Time: } THT = \max(0, TTRT - TRT) \mid \text{Efficiency } \eta = THT / TTRT$

1. Token Passing & Active Monitor

In Token Ring (IEEE 802.5), stations utilize a 3-byte circulating token frame. An Active Monitor prevents network deadlock by eliminating orphaned circulating frames via the Monitor Bit and regenerating lost tokens.

2. FDDI Dual Ring Self-Healing

FDDI Timed Token Equations

$$THT = \max(0, TTRT - TRT)$$

$$\text{Transmitted Asynchronous Bits} = THT \cdot 100 \text{ Mbps}$$

Upon a cable cut, adjacent Dual Attachment Stations (DAS) execute optical wrap-around, collapsing the primary and secondary counter-rotating rings into a single fault-tolerant operational ring.